PROJECT SPARK / LEVEL 15

Dual LED Parallel Circuit Challenge | 5 V | Five required objectives

Mission: power red and green LEDs in parallel, each with its own series resistor. Replace the incorrect 150 ohm resistor with 330 ohm, correct the green LED polarity, and check branch and source currents.

1. Required components

Part	Placement / value
PowerSupply_01 + Power_Switch	Existing 5 V supply and switch
R1 workstation	Initial 150 ohm; replace with shelf 330 ohm
LED1 / red	Existing SparkLED; positive terminal to resistor
R2 fixed	470 ohm resistor on parallel green branch
LED2 / green	New SparkLED; initially reverse wire connection
LCD / meter (optional)	Across R1 for voltage reading

2. Correct final topology

```
+5V -- Switch --+-- 330R -- Red LED ---+-- GND
| |
+-- 470R -- Green LED -+
```

Both LED anodes face their series resistors. Both cathodes return to Battery_negative. Do not wire either LED directly across 5 V.

3. Exact electrical connections

#	From	To
1	Battery_positive	SwitchInput
2	SwitchOutput	R1_Input
3	R1_Output	LED_Anode
4	LED_Cathode	Battery_negative
5	SwitchOutput	Resistor_470_Input
6	Resistor_470_Output	LED2_Anode
7	LED2_Cathode	Battery_negative
8 optional	LCD_Terminal_A	R1_Input
9 optional	LCD_Terminal_B	R1_Output

LEVEL 15 / UNITY BUILD PROCESS

4. Level prefab creation

1. Duplicate the working Level 14 layout as original Level_15_Layout prefab; do not duplicate the Level 14 definition asset over Level_15.asset.

2. Set SparkLevelLayout / loader references to the Level_15.asset definition and confirm Level 15 prefab is selected by your level loader.

3. Keep the 5 V power supply and switch; place the red LED, the fixed 470 ohm resistor and new green SparkLED.

4. Use the existing replaceable resistor workstation slot Register_Slot_2. The starting installed component is incorrect 150 ohm; correct shelf replacement is 330 ohm.

5. Configure correct anode/cathode terminal names and create stable IDs in the primary SparkLevelSceneBindings.

6. Save prefab. Import Level_15_DualLED_Import.txt into original Level_15.asset with existing Project Spark TXT importer.

7. Start Play Mode, make the seven required wires, replace R1 with 330 ohm and check all five objectives.

5. Primary SparkLevelSceneBindings mapping

Kind	Binding ID	Actual target
Fixed component	led	Red LED SparkLED (verify current prefab ID)
Fixed component	LED2	Green LED SparkLED
Fixed component	Resistor_470_Ω	Fixed 470 ohm SparkResistor
Fixed component	PowerSupply_01	Power supply
Fixed terminal	LED_Anode / LED_Cathode	Red LED anode / cathode
Fixed terminal	LED2_Anode / LED2_Cathode	Green LED anode / cathode
Fixed terminal	Resistor_470_Input / Output	470 ohm resistor terminals
Runtime component	R1_Component	Installed shelf resistor
Runtime terminals	R1_Input / R1_Output	Installed shelf resistor terminals

Critical: runtime IDs are declarations, not fixed object references. Add the three runtime IDs only under Runtime Component IDs and Runtime Terminal IDs, never the fixed scene binding arrays.

6. Workstation bridge

Set Slot ID = **Register_Slot_2**; Component Binding ID = **R1_Component**; Input / Output Binding ID = **R1_Input** / **R1_Output**; Electrical Type = **SparkResistor**; suffixes _**Input** / _**Output**. Confirm the input/output terminal suffixes match the shelf prefab. Use Synchronize + Report Now after installing 330 ohm.

LEVEL 15 / TEST & TROUBLESHOOT

7. Five completion targets in imported TXT

Target	Component ID	Pass criterion
Red LED Conducting	led	ComponentConducting = true
Green LED Conducting	LED2	ComponentConducting = true
Red Branch Current	R1_Component	8.0 - 10.5 mA
Green Branch Current	Resistor_470_Ω	5.5 - 7.0 mA (provisional)
Total Source Current	PowerSupply_01	14.0 - 17.0 mA (provisional)

8. Theoretical measurements

At nominal 5 V and approximately 2.0 V forward drop on BOTH LEDs: red (5 - 2)/330 = **9.09 mA**; green (5 - 2)/470 = **6.38 mA**; source = **15.47 mA**. These are design estimates, not verified simulator readings. LED model, wire/contact drops and a different green forward voltage alter them.

9. Test sequence

1. Check both LEDs have SparkLED components and that the target IDs resolve to those exact components.

2. Run with 150 ohm in R1: current-related objectives should fail. Turn power OFF.

3. Replace with 330 ohm in Register_Slot_2 and turn power ON.

4. Run read-only source-current audit: capture red current, green current and source current; confirm red + green approximately equals source.

5. Confirm both LED target logs report Passed=True and workstation diagnostics show Rule: VALID.

6. If green or source targets fail despite correct wiring, calibrate target ranges against real SparkLED forward voltage and re-export the definition.

10. Troubleshooting without solver modifications

LED target fails: verify exact fixed ID and SparkLED component reference; red LED ID **led** was copied from uploaded Level 14 asset, while **LED2** is a proposed new binding. A visible glowing LED alone does not prove evaluator binding resolution.

R1 current fails: check Runtime Component IDs = R1_Component; runtime terminals = R1_Input and R1_Output; bridge slot = Register_Slot_2; source item physically installed. Run Synchronize + Report Now.

Green/source current fails: audit physical values and adjust only definition thresholds if LED forward voltage differs from assumption. Keep fixed IDs case-sensitive.

Asset identity: package preserves Asset GUID `6284c7018244ffb4a842917d2c9049f8`, path and serialized field keys; the imported definition uses levelId **level_15**.